

LABSHEET 1

1. Write a Java program to display your name, age, and college name using variables of appropriate data types

```
public class StudentDetails {  
    public static void main(String[] args) {  
  
        // Declare variables  
        String name = "Rahul";  
        int age = 20;  
        String collegeName = "ABC College";  
  
        // Display student details  
        System.out.println("Name: " + name);  
        System.out.println("Age: " + age);  
        System.out.println("College Name: " + collegeName);  
    }  
}
```

Output

```
Name: Rahul  
Age: 20  
College Name: ABC College
```

2. Input two integers and print the result of all arithmetic operations (addition, subtraction, multiplication, division, and modulus).

```
import java.util.Scanner;  
  
public class ArithmeticOperations {  
    public static void main(String[] args) {  
  
        Scanner sc = new Scanner(System.in);  
  
        // Input two integers  
        System.out.print("Enter first integer: ");  
        int a = sc.nextInt();  
  
        System.out.print("Enter second integer: ");  
        int b = sc.nextInt();  
  
        // Arithmetic operations
```

```

        System.out.println("Addition: " + (a + b));
        System.out.println("Subtraction: " + (a - b));
        System.out.println("Multiplication: " + (a * b));
        System.out.println("Division: " + (a / b));
        System.out.println("Modulus: " + (a % b));

        sc.close();
    }
}

```

Output

```

Enter first integer: 20
Enter second integer: 6
Addition: 26
Subtraction: 14
Multiplication: 120
Division: 3
Modulus: 2

```

3. Write a Java program to check whether a given number is even or odd using the modulus operator and if-else

```

import java.util.Scanner;

public class EvenOdd {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        // Input a number
        System.out.print("Enter a number: ");
        int num = sc.nextInt();

        // Check whether the number is even or odd
        if (num % 2 == 0) {
            System.out.println(num + " is Even");
        } else {
            System.out.println(num + " is Odd");
        }

        sc.close();
    }
}

```

Output

```
Enter a number: 24
24 is Even
```

4. Write a program to demonstrate the use of relational operators by comparing two numbers.

```
import java.util.Scanner;

public class RelationalOperators {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        // Input two numbers
        System.out.print("Enter first number: ");
        int a = sc.nextInt();

        System.out.print("Enter second number: ");
        int b = sc.nextInt();

        // Demonstrate relational operators
        System.out.println("a == b : " + (a == b));
        System.out.println("a != b : " + (a != b));
        System.out.println("a > b : " + (a > b));
        System.out.println("a < b : " + (a < b));
        System.out.println("a >= b : " + (a >= b));
        System.out.println("a <= b : " + (a <= b));

        sc.close();
    }
}
```

Output

```
Enter first number: 15
Enter second number: 10
a == b : false
a != b : true
a > b : true
```

```
a < b   : false
a >= b  : true
a <= b  : false
```

5. Accept a floating-point number and convert it explicitly to an integer. Print both the original and the converted values.

```
import java.util.Scanner;

public class TypeCasting {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        // Input a floating-point number
        System.out.print("Enter a floating-point number: ");
        double num = sc.nextDouble();

        // Explicitly convert double to int
        int convertedValue = (int) num;

        // Display both values
        System.out.println("Original value: " + num);
        System.out.println("Converted integer value: " + convertedValue);

        sc.close();
    }
}
```

Output

```
Enter a floating-point number: 25.75
Original value: 25.75
Converted integer value: 25
```

6. Write a program to accept a character and print its ASCII value

```
import java.util.Scanner;

public class ASCIIValue {
    public static void main(String[] args) {
```

```

Scanner sc = new Scanner(System.in);

// Accept a character
System.out.print("Enter a character: ");
char ch = sc.next().charAt(0);

// Convert character to ASCII value
int asciiValue = (int) ch;

// Display the result
System.out.println("Character: " + ch);
System.out.println("ASCII value: " + asciiValue);

sc.close();
}
}

```

Output

```

Enter a character: A
Character: A
ASCII value: 65

```

7. Input a number and check if it is divisible by both 3 and 5 using logical operators.

```

import java.util.Scanner;

public class DivisibleBy3And5 {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        // Input a number
        System.out.print("Enter a number: ");
        int num = sc.nextInt();

        // Check divisibility by both 3 and 5
        if (num % 3 == 0 && num % 5 == 0) {
            System.out.println(num + " is divisible by both 3 and 5.");
        } else {
            System.out.println(num + " is not divisible by both 3 and 5.");
        }
    }
}

```

```
    }  
    sc.close();  
}  
}
```

Output

```
Enter a number: 30  
30 is divisible by both 3 and 5.\
```

8. Input a character and check whether it is a vowel or a consonant using if-else.

```
import java.util.Scanner;  
  
public class VowelConsonant {  
    public static void main(String[] args) {  
  
        Scanner sc = new Scanner(System.in);  
  
        // Input a character  
        System.out.print("Enter a character: ");  
        char ch = sc.next().charAt(0);  
  
        // Check whether the character is a vowel or consonant  
        if (ch == 'a' || ch == 'e' || ch == 'i' || ch == 'o' || ch == 'u' ||  
            ch == 'A' || ch == 'E' || ch == 'I' || ch == 'O' || ch == 'U') {  
            System.out.println(ch + " is a Vowel.");  
        } else {  
            System.out.println(ch + " is a Consonant.");  
        }  
  
        sc.close();  
    }  
}
```

Output

```
Enter a character: E  
E is a Vowel.
```

9. Input marks of three subjects, calculate total and percentage, and print pass/fail using if-else (passing criteria: $\geq 40\%$ in all subjects).

```
import java.util.Scanner;
```

```
public class MarksResult {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        // Input marks of three subjects
        System.out.print("Enter marks in Subject 1: ");
        double sub1 = sc.nextDouble();

        System.out.print("Enter marks in Subject 2: ");
        double sub2 = sc.nextDouble();

        System.out.print("Enter marks in Subject 3: ");
        double sub3 = sc.nextDouble();

        // Calculate total and percentage
        double total = sub1 + sub2 + sub3;
        double percentage = (total / 300) * 100;

        // Display total and percentage
        System.out.println("Total Marks: " + total);
        System.out.println("Percentage: " + percentage + "%");

        // Check pass or fail
        if (sub1 >= 40 && sub2 >= 40 && sub3 >= 40) {
            System.out.println("Result: PASS");
        } else {
            System.out.println("Result: FAIL");
        }

        sc.close();
    }
}
```

Output

Enter marks in Subject 1: 75
Enter marks in Subject 2: 68
Enter marks in Subject 3: 82
Total Marks: 225.0
Percentage: 75.0%
Result: PASS

10. Write a program to find the greatest of two numbers using if-else.

```
import java.util.Scanner;

public class GreatestOfTwo {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        // Input two numbers
        System.out.print("Enter first number: ");
        int a = sc.nextInt();

        System.out.print("Enter second number: ");
        int b = sc.nextInt();

        // Find the greatest number
        if (a > b) {
            System.out.println(a + " is the greatest number.");
        } else if (b > a) {
            System.out.println(b + " is the greatest number.");
        } else {
            System.out.println("Both numbers are equal.");
        }

        sc.close();
    }
}
```

Output

Enter first number: 25
Enter second number: 40

40 is the greatest number.

11. Input three numbers and print the largest using nested if-else

```
import java.util.Scanner;

public class LargestOfThree {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        // Input three numbers
        System.out.print("Enter first number: ");
        int a = sc.nextInt();

        System.out.print("Enter second number: ");
        int b = sc.nextInt();

        System.out.print("Enter third number: ");
        int c = sc.nextInt();

        // Find the largest using nested if-else
        if (a > b) {
            if (a > c) {
                System.out.println(a + " is the largest number.");
            } else {
                System.out.println(c + " is the largest number.");
            }
        } else {
            if (b > c) {
                System.out.println(b + " is the largest number.");
            } else {
                System.out.println(c + " is the largest number.");
            }
        }

        sc.close();
    }
}
```

Output

```
Enter first number: 25
Enter second number: 40
Enter third number: 30
40 is the largest number.
```

12. Accept a year and check whether it is a leap year using if-else.

```
import java.util.Scanner;

public class LeapYear {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        // Input year
        System.out.print("Enter a year: ");
        int year = sc.nextInt();

        // Check whether the year is a leap year
        if (year % 400 == 0) {
            System.out.println(year + " is a Leap Year.");
        } else if (year % 100 == 0) {
            System.out.println(year + " is not a Leap Year.");
        } else if (year % 4 == 0) {
            System.out.println(year + " is a Leap Year.");
        } else {
            System.out.println(year + " is not a Leap Year.");
        }

        sc.close();
    }
}
```

Output

```
Enter a year: 2024
2024 is a Leap Year.
```

13. Accept a number and check whether it is positive, negative, or zero using if-else.

```
import java.util.Scanner;
```

```

public class NumberCheck {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        // Input a number
        System.out.print("Enter a number: ");
        int num = sc.nextInt();

        // Check whether positive, negative, or zero
        if (num > 0) {
            System.out.println(num + " is Positive.");
        } else if (num < 0) {
            System.out.println(num + " is Negative.");
        } else {
            System.out.println("The number is Zero.");
        }

        sc.close();
    }
}

```

Output

```

Enter a number: 25
25 is Positive

```

14. Accept a character and check if it is a digit, uppercase letter, lowercase letter, or a special character using character classification and if-else.

```

import java.util.Scanner;

public class CharacterClassification {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        // Input a character
        System.out.print("Enter a character: ");
        char ch = sc.next().charAt(0);

        // Character classification using if-else
        if (ch >= '0' && ch <= '9') {
            System.out.println(ch + " is a Digit.");
        }
    }
}

```

```

    } else if (ch >= 'A' && ch <= 'Z') {
        System.out.println(ch + " is an Uppercase Letter.");
    } else if (ch >= 'a' && ch <= 'z') {
        System.out.println(ch + " is a Lowercase Letter.");
    } else {
        System.out.println(ch + " is a Special Character.");
    }

    sc.close();
}
}

```

Output

Enter a character: 7
7 is a Digit.

15. Accept a 3-digit number and check whether it is an Armstrong number using arithmetic operators and if-else.

```

import java.util.Scanner;

public class ArmstrongNumber {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int n, original, a, b, c, sum;

        System.out.print("Enter a 3-digit number: ");
        n = sc.nextInt();

        original = n;

        a = n / 100;
        b = (n / 10) % 10;
        c = n % 10;

        sum = (a * a * a) + (b * b * b) + (c * c * c);

        if (sum == original)
            System.out.println(original + " is an Armstrong number.");
        else
            System.out.println(original + " is not an Armstrong number.");
    }
}

```

```
        sc.close();
    }
}
```

Output

Enter a 3-digit number: 153
153 is an Armstrong number.

16. Input salary and calculate tax based on income slabs using if-else:
Income $\leq$ 250000: No tax
250001 to 500000: 5%
500001 to 1000000: 20%
Above 1000000: 30%

```
import java.util.Scanner;

public class SalaryTax {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        double salary, tax;

        System.out.print("Enter salary: ");
        salary = sc.nextDouble();

        if (salary <= 250000)
            tax = 0;
        else if (salary <= 500000)
            tax = salary * 0.05;
        else if (salary <= 1000000)
            tax = salary * 0.20;
        else
            tax = salary * 0.30;

        System.out.println("Tax = " + tax);

        sc.close();
    }
}
```

Output

Enter salary: 600000
Tax = 120000.0

17. Accept an integer and check whether it is a palindrome using arithmetic operations and type casting.

```
import java.util.Scanner;

public class Palindrome {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int num, original, remainder, reverse = 0;

        System.out.print("Enter an integer: ");
        num = sc.nextInt();

        original = num;

        while (num != 0) {
            remainder = num % 10;
            reverse = reverse * 10 + remainder;
            num = num / 10;
        }

        if (original == reverse)
            System.out.println(original + " is a palindrome.");
        else
            System.out.println(original + " is not a palindrome.");

        sc.close();
    }
}
```

Output

Enter an integer: 121
121 is a palindrome.

18. Accept a number and print whether it is prime or not using if-else.

```
import java.util.Scanner;

public class PrimeNumber {
```

```

public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);

    int n;
    boolean prime = true;

    System.out.print("Enter a number: ");
    n = sc.nextInt();

    if (n <= 1) {
        prime = false;
    } else {
        for (int i = 2; i <= n / 2; i++) {
            if (n % i == 0) {
                prime = false;
                break;
            }
        }
    }

    if (prime)
        System.out.println(n + " is a prime number.");
    else
        System.out.println(n + " is not a prime number.");

    sc.close();
}
}

```

Output

```

Enter a number: 17
17 is a prime number.

```

19. Accept a character and use if-else to identify whether it's a valid operator (+, -, *, /) and perform the corresponding operation on two input numbers.

```

import java.util.Scanner;

public class OperatorCheck {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        double num1, num2, result;
        char operator;
    }
}

```

```

System.out.print("Enter first number: ");
num1 = sc.nextDouble();

System.out.print("Enter second number: ");
num2 = sc.nextDouble();

System.out.print("Enter operator (+, -, *, /): ");
operator = sc.next().charAt(0);

if (operator == '+') {
    result = num1 + num2;
    System.out.println("Result = " + result);
}
else if (operator == '-') {
    result = num1 - num2;
    System.out.println("Result = " + result);
}
else if (operator == '*') {
    result = num1 * num2;
    System.out.println("Result = " + result);
}
else if (operator == '/') {
    if (num2 != 0) {
        result = num1 / num2;
        System.out.println("Result = " + result);
    } else {
        System.out.println("Cannot divide by zero.");
    }
}
else {
    System.out.println("Invalid operator.");
}

sc.close();
}
}

```

Output

```

Enter first number: 20
Enter second number: 5
Enter operator (+, -, *, /): +
Result = 25.0

```


20. Accept age and gender and determine eligibility for marriage (male ≥ 21 , female ≥ 18) using nested if-else conditions.

```
import java.util.Scanner;

public class MarriageEligibility {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int age;
        char gender;

        System.out.print("Enter age: ");
        age = sc.nextInt();

        System.out.print("Enter gender (M/F): ");
        gender = sc.next().charAt(0);

        if (gender == 'M' || gender == 'm') {
            if (age >= 21) {
                System.out.println("Eligible for marriage.");
            } else {
                System.out.println("Not eligible for marriage.");
            }
        }
        else if (gender == 'F' || gender == 'f') {
            if (age >= 18) {
                System.out.println("Eligible for marriage.");
            } else {
                System.out.println("Not eligible for marriage.");
            }
        }
        else {
            System.out.println("Invalid gender.");
        }

        sc.close();
    }
}
```

Output

```
Enter age: 22
Enter gender (M/F): M
Eligible for marriage.
```

21. Accept three angles and check whether they can form a valid triangle (sum of angles = 180°).

```
import java.util.Scanner;

public class ValidTriangle {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int angle1, angle2, angle3, sum;

        System.out.print("Enter first angle: ");
        angle1 = sc.nextInt();

        System.out.print("Enter second angle: ");
        angle2 = sc.nextInt();

        System.out.print("Enter third angle: ");
        angle3 = sc.nextInt();

        sum = angle1 + angle2 + angle3;

        if (angle1 > 0 && angle2 > 0 && angle3 > 0 && sum == 180) {
            System.out.println("The angles can form a valid triangle.");
        } else {
            System.out.println("The angles cannot form a valid triangle.");
        }

        sc.close();
    }
}
```

Output

```
Enter first angle: 60
Enter second angle: 60
Enter third angle: 60
The angles can form a valid triangle.
```

22. Input the length and breadth of a rectangle, and determine whether it is a square using if-else.

```
import java.util.Scanner;

public class SquareCheck {
```

```

public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);

    int length, breadth;

    System.out.print("Enter length: ");
    length = sc.nextInt();

    System.out.print("Enter breadth: ");
    breadth = sc.nextInt();

    if (length == breadth) {
        System.out.println("The rectangle is a square.");
    } else {
        System.out.println("The rectangle is not a square.");
    }

    sc.close();
}
}

```

Output

```

Enter length: 10
Enter breadth: 10
The rectangle is a square.

```

23. Input a temperature in Celsius and convert it to Fahrenheit using type casting and expressions.

```

import java.util.Scanner;

public class TemperatureConversion {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        double celsius, fahrenheit;

        System.out.print("Enter temperature in Celsius: ");
        celsius = sc.nextDouble();

        fahrenheit = (celsius * 9 / 5) + 32;

        System.out.println("Temperature in Fahrenheit = " + fahrenheit);
    }
}

```

```
        sc.close();
    }
}
```

Output

```
Enter temperature in Celsius: 25
Temperature in Fahrenheit = 77.0
```

24. Accept a number and check whether it is a perfect number (sum of divisors = number) using basic arithmetic and if-else.

```
import java.util.Scanner;

public class PerfectNumber {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int num, sum = 0;

        System.out.print("Enter a number: ");
        num = sc.nextInt();

        for (int i = 1; i <= num / 2; i++) {
            if (num % i == 0) {
                sum = sum + i;
            }
        }

        if (sum == num) {
            System.out.println(num + " is a perfect number.");
        } else {
            System.out.println(num + " is not a perfect number.");
        }

        sc.close();
    }
}
```

Output

```
Enter a number: 28
28 is a perfect number.
```

25. Accept two integers and a logical operator as a character. Use if-else to perform the logical operation and display the result.

```
import java.util.Scanner;

public class LogicalOperation {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int num1, num2;
        char operator;
        boolean result;

        System.out.print("Enter first integer: ");
        num1 = sc.nextInt();

        System.out.print("Enter second integer: ");
        num2 = sc.nextInt();

        System.out.print("Enter logical operator (&, |): ");
        operator = sc.next().charAt(0);

        if (operator == '&') {
            result = (num1 != 0) && (num2 != 0);
            System.out.println("Result = " + result);
        }
        else if (operator == '|') {
            result = (num1 != 0) || (num2 != 0);
            System.out.println("Result = " + result);
        }
        else {
            System.out.println("Invalid logical operator.");
        }

        sc.close();
    }
}
```

Output

```
Enter first integer: 10
Enter second integer: 20
Enter logical operator (&, |): &
Result = true
```

26. Input a five-digit number and check whether the first and last digits are the same using arithmetic expressions.

```
import java.util.Scanner;

public class FirstLastDigit {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int num, firstDigit, lastDigit;

        System.out.print("Enter a five-digit number: ");
        num = sc.nextInt();

        firstDigit = num / 10000;
        lastDigit = num % 10;

        if (firstDigit == lastDigit) {
            System.out.println("First and last digits are the same.");
        } else {
            System.out.println("First and last digits are not the same.");
        }

        sc.close();
    }
}
```

Output

```
Enter a five-digit number: 12321
First and last digits are the same.
```

27. Input the cost price and selling price of an item. Determine profit or loss, and by how much, using if-else.

```
import java.util.Scanner;

public class ProfitLoss {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        double costPrice, sellingPrice, amount;

        System.out.print("Enter cost price: ");
        costPrice = sc.nextDouble();
```

```

System.out.print("Enter selling price: ");
sellingPrice = sc.nextDouble();

if (sellingPrice > costPrice) {
    amount = sellingPrice - costPrice;
    System.out.println("Profit = " + amount);
}
else if (costPrice > sellingPrice) {
    amount = costPrice - sellingPrice;
    System.out.println("Loss = " + amount);
}
else {
    System.out.println("No profit, no loss.");
}

sc.close();
}
}

```

Output

```

Enter cost price: 500
Enter selling price: 650
Profit = 150.0

```

28. Accept a number and display the number of digits using expressions and if-else.

```

import java.util.Scanner;

public class NumberOfDigits {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int num, digits;

        System.out.print("Enter a number: ");
        num = sc.nextInt();

        num = Math.abs(num);

        if (num < 10) {
            digits = 1;
        } else if (num < 100) {
            digits = 2;
        }
    }
}

```

```

    } else if (num < 1000) {
        digits = 3;
    } else if (num < 10000) {
        digits = 4;
    } else if (num < 100000) {
        digits = 5;
    } else {
        digits = 6;
    }

    System.out.println("Number of digits = " + digits);

    sc.close();
}
}

```

Output

```

Enter a number: 12345
Number of digits = 5

```

29. Input the height (in cm) of a person and classify them as short (<150), average (150–170), or tall (>170).

```

import java.util.Scanner;

public class HeightClassification {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int height;

        System.out.print("Enter height in cm: ");
        height = sc.nextInt();

        if (height < 150) {
            System.out.println("Short");
        } else if (height <= 170) {
            System.out.println("Average");
        } else {
            System.out.println("Tall");
        }

        sc.close();
    }
}

```



```
}
```

Output

Enter height in cm: 145

Short

30. Write a Java program to simulate a simple calculator that takes two numbers and an operator (+, -, *, /) and performs the corresponding operation using if-else.

```
import java.util.Scanner;

public class SimpleCalculator {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        double num1, num2, result;
        char operator;

        System.out.print("Enter first number: ");
        num1 = sc.nextDouble();

        System.out.print("Enter second number: ");
        num2 = sc.nextDouble();

        System.out.print("Enter operator (+, -, *, /): ");
        operator = sc.next().charAt(0);

        if (operator == '+') {
            result = num1 + num2;
            System.out.println("Result = " + result);
        }
        else if (operator == '-') {
            result = num1 - num2;
            System.out.println("Result = " + result);
        }
        else if (operator == '*') {
            result = num1 * num2;
            System.out.println("Result = " + result);
        }
        else if (operator == '/') {
            if (num2 != 0) {
                result = num1 / num2;
                System.out.println("Result = " + result);
            } else {
```

```
        System.out.println("Cannot divide by zero.");
    }
}
else {
    System.out.println("Invalid operator.");
}

    sc.close();
}
}
```

Output

```
Enter first number: 20
Enter second number: 5
Enter operator (+, -, *, /): +
Result = 25.0
```

